

LAB – 4

GREEDY: KRUSKALS ALGORITHM

CODE

```
#include <stdio.h>

#define MAX 100

struct Edge {
    int u, v, weight;
};

int parent[MAX];

int find(int i) {
    while (parent[i] != i)
        i = parent[i];

    return i;
}

void unionSet(int u, int v) {
    int rootU = find(u);
    int rootV = find(v);

    parent[rootV] = rootU;
}

void sortEdges(struct Edge edges[], int e) {
    for (int i = 0; i < e - 1; i++) {
        for (int j = 0; j < e - i - 1; j++) {
            if (edges[j].weight > edges[j + 1].weight) {
                struct Edge temp = edges[j];
                edges[j] = edges[j + 1];
                edges[j + 1] = temp;
            }
        }
    }
}
```

```

int main() {

    int n, e;
    struct Edge edges[MAX];

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter number of edges: ");
    scanf("%d", &e);

    printf("Enter edges (u v weight):\n");

    for (int i = 0; i < e; i++) {
        scanf("%d %d %d",
            &edges[i].u,
            &edges[i].v,
            &edges[i].weight);
    }
}

```

```

for (int i = 0; i < n; i++) {
    parent[i] = i;
}

sortEdges(edges, e);

printf("\nEdges selected for MST:\n");

int count = 0;
int totalWeight = 0;

for (int i = 0; i < e && count < n - 1; i++) {

    int u = edges[i].u;
    int v = edges[i].v;

    if (find(u) != find(v)) {

        printf("%d -- %d = %d\n",

```

```
    if (find(u) != find(v)) {  
  
        printf("%d -- %d = %d\n",  
              u, v, edges[i].weight);  
  
        totalWeight += edges[i].weight;  
  
        unionSet(u, v);  
  
        count++;  
    }  
}
```

```
    if (count != n - 1) {  
  
        printf("\nMST cannot be formed.\n");  
        printf("The graph is disconnected.\n");  
  
    }  
    else {  
  
        printf("\nTotal weight of MST = %d\n", totalWeight);  
    }  
  
    return 0;  
}
```

OUTPUT:

```
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (u v weight):
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4

Edges selected for MST:
2 -- 3 = 4
0 -- 3 = 5
0 -- 1 = 10

Total weight of MST = 19
```